

Uncountable plurals as maximal clusters

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Abstract

English uncountable plurals are plurals whose parts cannot be counted directly, such as (**five*) *oats*/*grasslands*/*news*/*measles*/*tweezers*. These form a heterogeneous group in terms of semantic field and various morphosyntactic tests. Nevertheless, this paper shows that a unified analysis can account for the fact that their parts cannot be counted directly. In particular, it argues that uncountable plurals denote maximal clusters in a mereotopological sense. The maximality of these clusters causes a conceptualisation as wholes, which prevents access to the parts. We provide a descriptive typology of English uncountable plurals to illustrate their analysis as clusters connected through either overlap (Strong Connectedness), adjacency (External Connectedness), or Common Linkage, a connectedness relation first introduced here.

Keywords: plurality, clusters, numerals, countability, mereotopology, English

1 Introduction

This paper is concerned with English UNCOUNTABLE PLURALS: nouns that receive plural morphology and are often morphosyntactically plural, but which do not easily combine with numerals. Numeral modification may be entirely out (**five news*) or unusual (*#twenty-three oats*), optionally depending on the context (as in (1b) below). The concepts denoted by these nouns have the puzzling property that it is usually intuitively clear what their parts are (and hence why they are lexicalised as plurals), and yet it is not possible to count the parts. As can be seen from the examples in (1), these nouns constitute a heterogeneous group in terms of morphosyntax and semantics.

- (1) a. *Blitz the* **twenty* ***oats*** *in a food processor.*
- b. *We entered an area with* (**five*) *rolling* ***grasslands***.
- c. *The* (**three*) ***news*** *was bad.*
- d. *Folke got the* (**four*) ***measles***.
- e. *There are four* **(pairs of)* ***tweezers*** *on the table.*

Some of these nouns are the plural of a regular singular (*oats*; *grasslands*), others are derived from an adjective (*news*), and still others are pluralia tantum formed on a bound root (*measles*; *tweezers*). Most trigger plural agreement, but *news* does not.¹ There are nouns denoting mass-like substances (*oats*), spatial regions (*grasslands*), and concrete entities (*tweezers*), but also nouns with a more temporal structure

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¹For Acquaviva (2008: 16), the fact that *news* fails to trigger plural agreement is an argument not to treat it as a plural. We take the position that any noun ending in *-s* that can be related to a morpheme without *-s* should be viewed as a plural when it has the specific characteristics of internal structure that are discussed below. Also note that the relation between plural nouns and plural verb agreement is not straightforward. Mass-count nouns such as *furniture* and *footwear* are semantically plural but require singular agreement on the verb, while group nouns like *committee* and *team* can determine plural agreement in some varieties of

(*news*) and proper nouns, in the case of *measles* referring to a disease. Some of them can be modified by both *some* and *many* (*grasslands*; *tweezers*), others only by *some* (*oats*; *news*); still others by neither (*measles*).² We cast our net deliberately wide: the nouns in (1) all morphologically reflect the fact that they denote a plurality.

It is not immediately clear what prevents modification by numerals for the nouns in this group. In view of its heterogeneity, this may not even be the same property for all nouns in the first place. In this paper we propose a semantic typology of uncountable plurals: nouns that have morphological (and often morphosyntactic) signs of denoting a plurality, but whose parts cannot normally be counted using modification by a numeral. Analysing the various types, we suggest that these nouns denote clustered pluralities: sets of entities which are clustered into a coherent whole. The fact that they denote sets motivates their plural morphology (and optionally agreement). Furthermore, we suggest that uncountable plurals specifically denote MAXIMAL CLUSTERS. Maximal entities are seen as “wholes”, and therefore it is not straightforward to access their parts. We argue that this explains the uncountability of these plurals.

The remainder of this paper is structured as follows. In section 2 we provide the necessary theoretical background for our analysis. Section 3 walks through the various types of uncountable plurals, after which section 4 defines the notion of maximal clusters and explains why they are uncountable. Section 5 concludes.

2 Theoretical background

The notion of clustered pluralities stems from mereotopology, for which we will follow Grimm (2012). In this approach, general extensional mereology is extended with the topological relation of Connectedness. Following standard mereological approaches we assume a parthood relation $x \leq y$ from which the notion of Overlap can be derived:

$$(2) \quad O(x, y) =_{\text{def}} \exists z [z \leq x \wedge z \leq y]$$

A mereological sum $x \oplus y$ can be defined as the set (a) that contains both x and y and (b) of which each member overlaps with x , y , or both. Notably, mereological sums say nothing about wholeness or individuation: two halves of a sphere can be summed to yield the whole sphere, but just so my finger and the lamp on my desk have a mereological sum. Connectedness can help define wholes: “two entities are connected if they share a common boundary” (Grimm 2012). Specifically, wholes can then be defined as Maximally Strongly Self-Connected entities. First, Self-Connectedness says that an entity cannot be divided into two unconnected parts:

$$(3) \quad SC(x) =_{\text{def}} \forall y, z [\forall w [O(w, x) \leftrightarrow (O(w, y) \vee O(w, z))] \rightarrow C(y, z)]$$

This distinguishes the sphere formed by two halves from the sum of my finger and the lamp of my desk. However, the sum of two entities that just “touch”, such as Belgium and the Netherlands, is still seen as Self-Connected. Strongly Self-Connectedness rules out such sums:

$$(4) \quad SSC(x) =_{\text{def}} SC(x) \wedge SC(ix) \\ \text{where } ix =_{\text{def}} \bigoplus \{y : IP(y, x)\} \quad (\text{interior}) \\ IP(x, y) =_{\text{def}} x \leq y \wedge \forall z [C(z, x) \rightarrow O(z, y)] \quad (\text{Internal Part})$$

English. Many uncountable plurals are based on the pluralisation of an adjective, at least some of which trigger singular agreement as well (*blues*, *narrows*, *news*). It is not our intention here to provide a full analysis of the factors determining the relation between semantically plural nouns and verb agreement.

²Roughly, it seems that *many* requires individuated parts; it can for instance modify *scrapings*, which can be related to separate events, but not *ashes*. *Some*, by contrast, requires that the cluster can be divided into separate portions, without requiring that these portions are clearly individuated. Thus, it can modify *ashes*, but not *Bahamas*: *ashes* can be divided into heaps, but the submerged base linking all islands in the *Bahamas* is an integral part of the concept (this is why a boat can *sink in the Bahamas*, which necessarily does not happen on one of the islands). We do not work this out further here.

The interior ix of x is the sum of all parts that are strictly internal to x (in the sense that anything connected to it overlaps with x). Strongly Self-Connectedness requires that an entity’s interior is Self-Connected; the sum of Belgium and the Netherlands is excluded because the interiors of these two countries are not connected. However, the sum of two thirds of a sphere is still Strongly Self-Connected, even though we would not want to see it as a whole. To distinguish these entities, Maximally Strongly Self-Connectedness relative to a property P only holds for the largest SSC entity satisfying P :

$$(5) \quad \text{MSSC}(P)(x) =_{\text{def}} P(x) \wedge \text{SSC}(x) \wedge \forall y [P(y) \wedge \text{SSC}(y) \wedge O(y, x) \leftrightarrow y \leq x]$$

As explained by Grimm, MSSC is a useful substitute for the property of atomicity in typical accounts of count nouns, since it sidesteps problems with concepts like *fence* (which can consist of smaller fences).

The notion of clusters is introduced to deal with aggregates like *sand* and collective nouns in languages with a collective–singulative opposition. These types of nouns can both refer to three mereotopologically different types of entities. First, a mass-like pile of sand counts as *sand*, but so does an individual (MSSC) grain of sand, and a plurality of such grains – in this *sand* is different from *water* which does not have minimal parts. In Welsh, which has a collective–singulative opposition, these three different types of entities are morphologically distinct. Aggregates like *tywod* ‘sand’ as well as entities which usually come in groups can be lexified as unmarked collectives (e.g., *chwain* ‘fleas’). These have a marked singulative form (*chwannen* ‘flea’) which itself has an inflectional plural (*chwannennau* ‘several fleas’). So, the analysis is that collectives denote clusters, singulatives denote MSSC entities, and the inflectional plural denotes sums of MSSC entities (as in English). Formally, clusters have the property that every pair of entities in the cluster is connected, possibly through other intermediate entities (e.g., two grains of sand in a pile need not touch, but need to be connected through other grains of sand):³

$$(6) \quad \text{Cluster}(P, C)(x) =_{\text{def}} \exists Z [x = \bigoplus Z \wedge \forall z, z' \in Z, \exists Y \subseteq Z [\text{TransitiveC}(P, C)(z, z', Y)]]$$

where $\text{TransitiveC}(P, C)(x, y, Z = \langle z_1, \dots, z_n \rangle) =_{\text{def}}$
 $\forall_{1 \leq i \leq n} [P(z_i)] \wedge x = z_1 \wedge y = z_n \wedge C(z_1, z_2) \wedge \dots \wedge C(z_{n-1}, z_n)$

The Cluster property allows for multiple “varieties” or “degrees of connectedness” C : Strong, External, and Proximate Connectedness. Strong Connectedness obtains between two entities whose interiors overlap, such as France and the Pyrenees. External Connectedness obtains between two connected entities whose interiors are unconnected, such as France and Spain (or touching grains of sand). Finally, Proximate Connectedness obtains between two entities if the distance between them is lower than some predicate-dependent threshold, as in a swarm of bees.

$$(7) \quad \text{StrongC}(x, y) =_{\text{def}} \exists w, z [w \leq x \wedge z \leq y \wedge \text{SSC}(w \oplus z)]$$

$$(8) \quad \text{ExtC}(x, y) =_{\text{def}} C(x, y) \wedge \neg C(ix, iy)$$

$$(9) \quad \text{ProxC}(x, y) =_{\text{def}} d(x, y) \leq n$$

The notion of cluster has been applied to a variety of phenomena, such as the distribution of singulative morphemes in languages like Welsh (Grimm 2012) and Ukrainian (Wagiel & Shlikhutka forthcoming), the distribution of the French quantifier *quelques* ‘a few’ (Gréa 2023), and *swarm* nouns (Henderson 2017). Notably, these phenomena can be linked to uncountability: Welsh and Ukrainian collectives cannot be counted, although the plural of the derived singulative can, as shown in (10) for Ukrainian. Although French *quelques* can modify countable pluralities (11a), it can also modify pluralia tantum, which cannot be modified by numerals or other quantifiers (11b). And to count the members of a *swarm*, one must use periphrasis rather than modifying the noun directly (*a swarm of eight* rather than **eight swarm*).

$$(10) \quad \text{Ukrainian} \qquad \qquad \qquad (\text{Wagiel \& Shlikhutka forthcoming})$$

³This definition differs slightly from Grimm’s version. We follow Wagiel (2021: 193 nn. 9–10) in restricting Y to subsets of Z , and requiring Z to be an ordered sequence, to avoid some undesirable consequences.

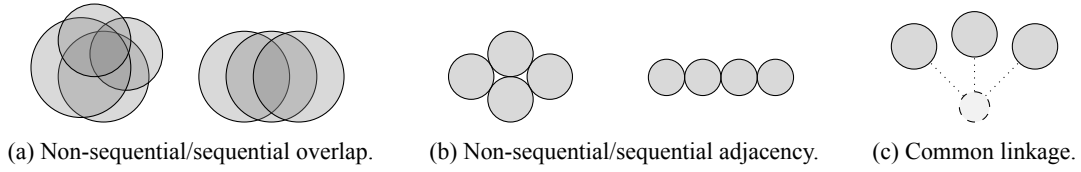


Figure 1: Different types of connectedness relations.

- a. * *dvi trav-y*
two grass-PL
Intended: ‘two grasses’
- b. *dvi trav-yn-y*
two grass-SGV-PL
‘two grass blades’
- (11) French (Gréa 2023: 430)
- a. *J’ai vu quelques/plusieurs chats.*
‘I have seen a few/several cats.’
- b. *Prenez donc *plusieurs/quelques épinards.*
‘Take several/some spinach(PL).’

Thus, there is good reason to believe that at least in some languages, numerals cannot modify NPs denoting clusters. In section 4, we will argue that this is because clusters are maximal: maximal clusters can be seen as “wholes”, which prevents easy access to their parts. Before we introduce the notion of maximal clusters, however, we show that English uncountable plurals can be analysed as clusters.

3 Descriptive typology

In order to show that English uncountable pluralities are clusters, we present a semantic typology of these nouns. Our data set consists of nouns marked as pluralia tantum and/or never modified by numerals in the Universal Dependencies data set of the English Web Treebank (Silveira et al. 2014), as well as nouns found in the literature on pluralia tantum and uncountability or through snowball sampling in dictionaries and personal reading.

The variables in the typology are as follows. First, we categorise pluralities as clusters of entities that are connected (in the mereotopological sense) through either Strong Connectedness (7) or External Connectedness (8).⁴ For the purpose of intuitive exposition, we will call these **OVERLAP** and **ADJACENCY**, respectively.⁵ In both types, the parts of the plurality may optionally be **SEQUENTIALLY ORDERED**. For instance, *depths* are sequentially overlapping (each deep region is deeper than the previous, with which it overlaps), while *sands* are non-sequentially overlapping (sandy regions may overlap with more than two other sandy regions, not forming a sequence). And the parts of *stairs* are sequentially adjacent (each step is higher/lower than the previous, which it only touches), whereas the parts of (*head*)*quarters* are non-sequentially adjacent (rooms or buildings may be adjacent to more than two other rooms or buildings, not forming a sequence). These connectedness relations are schematised in figs. 1a and 1b.

Besides overlap and adjacency we also recognise another type of connectedness, which we call **COMMON LINKAGE**. In this type, the parts of the plurality do not touch each other, but there is an implicit

⁴We do not have examples that require Proximate Connectedness (9), a point to which we will return in section 5.

⁵This use of “overlap” is stronger than the mereological notion of the same name, which we will not use in this article.

	Overlap		Adjacency		Common Linkage
	Non-sequential	Sequential	Non-sequential	Sequential	
Primarily spatial	<i>sands, Cotswolds, bangs</i>	<i>depths, heavens, intestines</i>	<i>muscles, (head)quarters, Netherlands, tweezers, pants</i>	<i>stairs</i>	<i>Bahamas, measles, chives</i>
Primarily temporal	<i>thoughts, blues, creeps</i>	<i>news, musings, ruminations, jitters, shakes</i>		<i>(his/the) eighties, directions, chord changes</i>	<i>hiccups, groceries, ashes, embers, scrapings, sweepings</i>
Primarily functional	<i>academics, tariffs, provisions</i>		<i>clothes, linguistics, details, particulars</i>		<i>buyables, brains, rights, notes (of peach)</i>

Table 1: Representative examples of spatial, temporal, and functional overlap and adjacency.

entity to which all can be linked. For instance, the spots through which *measles* presents itself are not overlapping or adjacent. However, they are seen as clustered together because they can all be linked to a common root cause. Sequentiality does not apply to common linkage; there is only non-sequential common linkage. Common Linkage is depicted in fig. 1c and can be formally defined as in (12), where *l* is the linking entity and *R* is the relation of parts of the plurality to this entity.⁶

$$(12) \text{ CommonLink}_{C_{l,R}}(x, y) =_{\text{def}} R(x, l) \wedge R(y, l)$$

Second, we recognise that these types of connectedness relations occur in different semantic domains: the SPATIAL, TEMPORAL, and FUNCTIONAL DOMAIN. For instance, *grasslands* are spatially connected but *news* is temporally connected. The parts of a particular plurality may be connected in multiple ways: for instance, the *Netherlands* are not only spatially but also functionally adjacent, since they are grouped in a single legal entity. The recognition of these three domains is in line with much cognitive linguistic work on the mapping of the spatial domain onto more abstract domains (e.g., Lakoff & Johnson 1980; Churchland 1986; Talmy 2000) as well as formal work in the mereological tradition (among many others, Bach 1986; Krifka 1989, 1992; Landman 2000; Link 1998 on events and Krifka 1996; Lahiri 2002; Schmitt 2019 on more abstract entities). Note that although Grimm’s (2012) examples assume a spatial understanding of mereotopological connectedness, nothing depends on this (in fact, connectedness can be any reflexive and symmetrical relation that is implied by mereological parthood and overlap). Grimm already suggests that future work can look at extensions to the framework for eventive and abstract nouns (2012: 601), and such extensions have already been explored in full formal detail by other scholars (Mazzola 2019; Wągiel 2021, 2024, 2025; McNally 2024).

These variables – type of connection (including sequentiality) and domain – are fully orthogonal: we find examples of all combinations (spatially adjacent, temporally overlapping, etc.). Intuitive examples of the various combinations are provided in table 1. We return to the gaps in this table below.

We should stress here that, as far as our data is concerned, these variables are only intended to describe the semantic space of uncountable pluralities: none of these variables has a reflex in the grammar of

⁶Whereas entities in a sequentially adjacent cluster, for instance, do not need to be adjacent (they only need to be *transitively* adjacent; see (6) and fig. 1b), this definition ensures that all entities in a cluster connected through common linkage are linked through the same linking entity.

English uncountable pluralities, as far as we are aware.⁷ It is the notion of cluster that has a grammatical reflex (preventing modification by numerals), and we only use types of connection and domain to show that the notion of cluster can account for the full breadth of uncountable pluralities.

The following subsections will systematically work through the spatial, temporal, and functional domain.

3.1 The spatial domain

Spatial overlap is most clearly represented by terms for geographical areas such as *sands*, *tropics*, *grass/heath/wet/...lands*, (*international*) *waters*, *deeps*, and *heavens*. These often but not always alternate with a corresponding singular which offers a different conceptualisation of the same real-world entity (*heaven* – *heavens*, but not **tropic* – *tropics*). For some of these, the overlapping regions constitute a (typically vertical) sequence: *deeps*, *depths*, and *heavens* are overlapping regions stacked on top of each other. Plural proper names for regions of non-individuated entities such as woods and mountains also reflect spatial overlap (*Cotswolds*, *Himalayas*, *Alps*, *Pyrenees*, ...). Whether sequential or not, overlapping geographical regions classify as clusters with a straightforward physical notion of overlap.

Some body parts also reflect spatial overlap. *Bangs* ‘hair hanging over the forehead’ does not refer to individual hairs but to sections of hair that fall together. These sections are non-individuated and overlapping, and also constitute a mereotopological cluster in this sense. The parts of *bowels*, *entrails*, *innards*, and *intestines* are not individuated either, but unlike *bangs* these body parts are sequentially clustered to the extent they are conceptualised as something through which food passes.

Other body parts are more clearly individuated; they form clusters connected by adjacency. This is the case for *muscles*: each muscle is well-individuated, but they are grouped together in the motor system. Another small set of spatially adjacent clusters is formed by cereals that are typically consumed in a porridge, namely *oats*, *grits*, and *groats*. They form a highly cohesive structure of partly disintegrated grains bound together by a starch network; as such, their parts are spatially adjacent.

We also categorise symmetrical entities such as *tweezers* under the notion of spatial adjacency. These mainly fall into two groups: tools such as *tweezers* (*scissors*, *handcuffs*, *pliers*, ...), and items of clothing or accessories, broadly construed (*pants*, *trousers*, *jeans*, *glasses*, *spectacles*, ...). Symmetrical entities with plural marking thus mostly relate to either paired body parts or to the pair consisting of the thumb and one or more other fingers. Only *gallows* and *narrows* ‘strait connecting two bodies of water’ are not related to the human body.⁸ In English, spatial adjacency is crucial for uncountability: *two shoes/socks/gloves* can be counted, but **two pants/spectacles/ear muffs* cannot.

As mentioned above, *stairs* is an example of sequential adjacency: the steps form a sequence and only touch each other (in contrast to *heavens*, which consists of overlapping regions).⁹

In some cases, parts are not connected physically, or the physical point of connection is invisible. This is the case for three main subgroups. We have already mentioned *measles*, which presents itself in spots that are neither overlapping nor adjacent. We argue that they are clustered through common linkage, because each spot is related to the same root cause. Other diseases, such as *haemorrhoids/piles*, *hives*, and *shingles*, can be analysed in the same way.¹⁰

⁷However, these variables do have a grammatical reflex in other languages. For instance, spatial and temporal sequentiality is marked by classifiers in certain languages (Aikhenvald 2025: 234–238), and Wagiel (2021) shows that Slavic has distinct derivational suffixes for spatially and functionally connected collectives. Henderson (2017) observes differences between spatial and temporal collective nouns (e.g., *swarm*) on the one hand and other collectives (e.g., *committee*) on the other (though note that some but not all have been argued to be due to differences in animacy instead; see Wagiel 2021: 179–182).

⁸*Gallows* were originally typically supported by two upright posts, hence its inclusion in this list of symmetrical entities.

⁹Strictly speaking, symmetrical entities like *tweezers* are sequential as well, but it is not very meaningful to talk about sequentiality in the context of plurals with less than three parts.

¹⁰There is one outlier in this group that is harder to account for: *rickets* ‘a condition that results in weak or soft bones in children and may have either dietary-deficiency or genetic causes’. As a loan from Greek ραχίτης, the -s should probably not be taken as an indicator of plurality.

Second, proper names of archipelagoes are often plural (*Antilles*, *Antipodes*, *Azores*, *Bahamas*, *Cyclades*, *Falklands*, *Hebrides*, *Philippines*, ...). Archipelagoes can form under various circumstances, but have in common that they have a shared base which is elevated compared to the surrounding body of water. Thus, the *Bahamas* are connected in the same way as *measles*, all parts being linked to the shared elevated base (in some cases, such geographical entities are also political entities, which we take to be a particular subcase of adjacency in the functional domain, discussed in section 3.3 below).

Third, plants with multiple blades or leaves growing from the same root structure often receive plural marking in a variety of languages, although languages typically use plural marking for only a few such plants, and make different choices as to which: English *chives*, (*salad*) *greens*; French *épinards* ‘spinach’; Dutch *raapstelen* ‘rapini’; Italian *cime di rapa* ‘id.’; etc. The common denominator in these plants is that the leaves are connected to the same growth point below the surface. Latin *aquae* ‘spring water’, though not a plant, also denotes multiple places where something springs from the earth, which are connected to the same source underground.

3.2 The temporal domain

As is often the case, what applies to space extends to time. The mereotopological notions of overlap (strong connectedness) and adjacency (external connectedness) map onto the temporal domain, and both have sequential and non-sequential subtypes.

Pure temporal overlap occurs in only a few nouns, such as *thoughts*, *ruminations*, *contemplations*, and *musings*: these consist of subevents which cannot be individuated but flow into each other. Most of these are sequential. For instance, the Cambridge Dictionary describes *musings* as “your thoughts or comments on something you have been thinking about carefully and for a long time”, emphasising sequential progress. *News*, *profits*, *savings*, and *dues* denote aggregates that accumulate over time in possibly overlapping but sequentially ordered events. *News* and *finances* combine temporal relations with functional ones, to be discussed in section 3.3.

Other nouns show non-sequential temporal overlap: the sensation of *having feelings for* someone or *mixed feelings about* something involves multiple feelings in one and the same moment. They are hard to individuate but clearly similar, contributing to a single complex experience. In that sense, they form a non-sequential cluster through temporal overlap. A number of emotional states share this property: *blues*, *creeps*, *doldrums*, *hackles*, *heebie-jeebies*, *hots* (in *have the hots for someone*), *jim-jams*, *runs*, *shits*, (*high*) *spirits*, *squirts*, and *willies*. However, when terms for emotional states point to repetitive events (as in *jitters* and *shakes*, both designating a state of nervousness), they are best seen as involving sequential rather than non-sequential overlap (since the individual jitters or shakes are not individuated, we still see them as temporally overlapping rather than adjacent).

Rains and *snows* can be both sequentially and non-sequentially overlapping. When a farmer talks about when *the first rains* arrive on her land, these follow each other and are sequentially overlapping. We see them as overlapping rather than adjacent because two rain showers may be interspersed by drizzle, so that we can also count them as a single shower. They are sequential because, no matter the level of granularity, the showers are ordered sequentially. However, in *the country has seen many rains this month*, a shower in one region may be contemporaneous with but distinct from several showers in another region. In such a situation, the showers cannot be ordered sequentially, and form a non-sequential cluster.

The clearest cases of temporal adjacency are found with time intervals: *the eighties* and *his eighties* both refer to ten successive years, connected only by the turn of the year. These are clearly sequential. *Chord changes*, which constitute a melodic pattern, are likewise sequentially adjacent. Sequential temporal adjacency combines with spatial sequential adjacency in (*asking*) *directions* and *whereabouts*. For instance, *whereabouts* denotes a cluster of time–location pairs: in an example such as *Can you account for your whereabouts between 8 and 10pm?*, a temporally adjacent sequence of overlapping or adjacent locations is requested. As we will discuss in more detail in section 3.3, sequential temporal adjacency also

combines with functional relations: in (*peace*) *talks* the talks are sequentially ordered but also related to each other in terms of content, a kind of functional adjacency; and *banns* and *nuptials* designate internally complex ceremonies the parts of which functionally complement each other.

Note that we do not find cases of non-sequential temporal adjacency. This seems reasonable given the linear nature of time.

Some uncountable pluralities, like *hiccups*, appear to refer to temporally sequential subevents, but are not overlapping or adjacent because there is time in between these subevents. These therefore require a different analysis, which the notion of common linkage can provide.¹¹ This renders the analysis of *hiccups* similar to that of *measles*, a desirable consequence: whereas the latter presents itself in multiple non-overlapping and non-adjacent spatial locations, *hiccups* presents itself in multiple non-overlapping and non-adjacent events. The same is true for *bends* ‘joint pain due to decompression sickness’, *chills* ‘feeling of coldness during a fever’, and *mumps* (which involves swelling around the sides of the face, causing grimaces). As with diseases with spatially distributed symptoms, we see these diseases as clustered because each instance of a symptom has the same root cause. Although the symptoms present themselves sequentially, they are not related to each other in a sequence: the first hiccup has the same relation (of being caused by the underlying ailment) to the second hiccup as to the third. These are therefore cases of temporal common linkage.

Temporal common linkage also occurs in uncountable plurals whose parts are physical entities which are the result of a single event that affects the nature or function of their source. As an example, consider *ashes*: this plural is typically used to refer to what is left after cremating or burning a single entity like a person or city. By contrast, a chimney can be clogged by the *ash* resulting from burning coal or wood over time; in this context, where the ash is built up in separate, unrelated events, it is odd to use plural *ashes*. Thus, *ashes*, *embers*, etc. denote entities or portions that are related to each other through the event in which they were created. For some of these the common link may be both temporal and spatial (e.g., *ashes* are the result of burning a single entity in a single event, but this requirement is not as strict for *embers*), and the resulting plurality may also be characterised by spatial overlap. For some of these event-related plurals, the plural is derived from a verb describing the change, as in *sweepings*, *dregs*, *grounds*, *leftovers*, and *leavings*. In other cases the event is just presupposed, as in *ashes* and *bones*.

There is quite some internal variation in this class of event-related uncountable plurals. *Ashes* and *embers* are the result of destruction, and so are *bones* and *remains*.¹² This is also true for *ruins*, and *vestiges*, but here the parts making up the plural seem to also retain or refer some of the internal spatial structure of the original individual, so that they are connected through temporal common linkage (through the event in which they were created), spatial common linkage (through the original entity), and spatial adjacency (of the original entity).¹³ *Fumes* and *wastes* are the result of an event separating them from their source. *Dregs* (of a liquid, or figuratively of a virus), (*coffee*) *grounds*, and *residues* are what remains after a liquid is removed from a container, and are also spatially overlapping. *Sediments*, *settlings*, *lees* ‘sediment of wine in a barrel’, and *deposits* ‘geological accumulations of natural matter’ involve sequential spatio-temporal overlap, as they are the result of depositing layers one after the other over time, rather than removing liquid. *Scrappings*, *sweepings*, *leavings*, and *leftovers* refer to accumulated small pieces with a

¹¹Grimm’s (2012) Proximate Connectedness relation – introduced for animals in swarms and the like – would also be able to capture *hiccups* in this way. However, we believe that the notion of Proximate Connectedness is not necessary after introducing connectedness in the functional domain, a point to which we return in the conclusion.

¹²These can be compared to singular *skeleton* on the one hand and countable *relics* on the other. *Skeleton* is singular because the parts are too dissimilar: a skull is clearly different from a shoulder blade, and these need to be positioned correctly to form a skeleton. In *bones* and *remains* the parts are seen as more homogeneous, allowing for a plural. This is also why bones and remains may be partial and incomplete, while *skeleton* presupposes completeness. However, they still constitute an event-based cluster, preventing countability. On the other side of the spectrum, *relics* each have a different origin story and transmission history; they do not form an event-based cluster and therefore are countable.

¹³*Ruins* has an interesting use in which there is no temporal common linkage: the *ruins of Troy* are the ruins of a city which was destroyed and rebuilt multiple times. They are not created in a single event, but are still linked through the same spatio-conceptual entity.

similar origin and shape that are the result of removing paint, dust, or food, respectively. *Minutes* and *proceedings* are created as the result of a single meeting or conference (their parts are also related in the functional domain, to be discussed below). Finally, *groceries* are related through a single shopping event, and *damages* ‘monetary compensation’ are related through a previously caused injury or loss. Although the specific type of event is different in all these cases, what unites them is a strong before-and-after relationship that seems central to their semantics, and which can be analysed as common linkage through an event.

3.3 The functional domain

We have already alluded to the functional domain as a secondary way in which the parts of some pluralities are clustered. Archipelagoes like the *Bahamas*, for instance, are spatially adjacent but also form political entities. These political entities are typically also internally complex, with lower governments responsible for each island or group of islands. These elements are connected in a functionally adjacent cluster: their functions are not overlapping (each being responsible for a different region) but highly similar (since the regions are adjacent to each other), and together they constitute a greater whole. This is not only true for archipelagoes. The *United States* and, historically, the *Netherlands* are also entities with a geographical and a political subdivision. The functional domain is not merely political; a similar structure can be recognised in (*head*)*quarters*, where the functions of the different rooms or buildings are distinct but complementary.

The temporal domain also interacts with the functional domain. *Peace talks*, *banns*, and *nuptials* all consist of sequentially adjacent subevents, but these subevents also have a functional relation to each other. The discussions in *peace talks* build on each other with the aim of culminating in a peace agreement. And the parts of *banns* and *nuptials* complement each other to achieve a religious goal. Both of these are forms of sequential functional adjacency. The parts of *minutes* and *proceedings* are related through the event in which they are created (non-sequential temporal adjacency) but also functionally by covering similar and/or complementary topics (non-sequential functional overlap or adjacency, respectively). *News* is produced in sequentially overlapping events and covers topics related to each other (non-sequential functional overlap). A project’s finances are hierarchically structured into subbudgets (non-sequential functional adjacency) but also has a corresponding planning (sequential temporal adjacency).

We can add here the special category of game names that are based on the terms for the objects that are played with (*billiards* – likely deriving from the French terms for either ‘stick’ or ‘ball’, *cards*, *checkers*, *craps*, *darts*, *dominoes*, *draughts*, *fives* – referring to either the number of fingers or the number of players on each team). These show functional adjacency in that the objects of the competing players have complementary roles, as well as sequential temporal adjacency since the players take turns moving them.

Functional overlap and adjacency often piggy-back on overlap or adjacency in the spatial or temporal domain. This explains why, although most countries have a division into lower governments, this functional clustering is only reflected in their name when it is based on a clearly pre-existing spatial division, as in the case of the *United States* and the *Netherlands*. However, there are also uncountable plurals that denote clusters bound together by purely functional relations. For instance, the *details* or *particulars* of a plan are functionally (non-sequentially) adjacent in that they complement each other in the larger whole: the function of each detail or particular piece of information is distinct but closely related to that of the others. Similarly, different *clothes* have the partly overlapping, partly adjacent functions of covering different body parts.¹⁴ Someone’s *credentials* or *qualifications*, like diplomas and previous experience, complement each other in showing that they are qualified for a job; they are functionally adjacent or overlapping depending on the extent to which they provide evidence for similar skills. Information bearers like *acts*, *annals*, and *archives* can also be uncountable. They form clusters related through functional overlap

¹⁴Although clothes are spatially adjacent when worn, they do not necessarily form a spatial cluster. For this reason we take the uncountability to be due to functional clustering.

and/or adjacency depending on the degree to which they are structured (i.e., whether the individual items bear information on overlapping or only related topics). Some of these concepts are additionally related through sequential temporal adjacency (for instance, *annals* correspond to consecutive periods of time).

There are four larger groups of nouns that are categorised by purely functional clustering. The first are nouns on *-ics*, such as *academics*, *acoustics*, *aerodynamics*, *athletics*, *civics*, *classics*, *dynamics*, *economics*, *electronics*, *ethics*, *genetics*, *linguistics*, *logistics*, *mathematics*, *mechanics*, *metaphysics*, *physics*, *politics*, *pragmatics*, *robotics*, *semantics*, and *statistics* (we can add to this list cases without *-ics*, such as *social studies*). Most of these have two readings: without a complement they refer to an academic discipline (*a professor of aerodynamics*); with a complement they refer roughly to the aspects of the complement qua that discipline (*the aerodynamics of the car*). Many of these disciplines can be analysed as having complementary subdisciplines. Thus, the plural *-s* in *linguistics* can be seen as motivated by the functionally adjacent fields of phonology, morphology, syntax, etc.; similarly, *mathematics* is made up of geometry, algebra, number theory, etc. This extends to many examples (*athletics*, *classics*, *social studies*, etc.). In other cases, the plural seems to be motivated by a complex structure of complementary entities or processes. Thus, *the electronics (in a computer)* refers to components like resistors and capacitors that are connected in a complex circuit, a type of functional adjacency; *the logistics of the Roman army* refers to sections and processes responsible for the movement of troops and supplies, which are likewise functionally connected; and *the dynamics of life* can be analysed similarly, if at a more abstract level. Interestingly, these readings differ in their agreement patterns. The discipline reading triggers singular agreement (*electronics is fascinating*), whereas the complementary structure reading triggers plural agreement (*these electronics are state-of-the-art*). This allows for a comparison with more well-known examples of hybrid nouns, such as *committee*.¹⁵

The second group consists of nouns on *-ables* denoting clusters with common linkage, the parts of these pluralities being linked only through having a particular property. *Buyables* are any items that can be bought – each buyable also has other functions but they are clustered together by virtue of being buyable. The same goes for *adjustables*, *burnables*, *combustibles*, *compostables*, etc., as well as some with a non-verbal root: *fashionables*, for instance, share the function of making someone conform to current fashion. *Wearables* are broader than *clothes*, including accessories like jewellery and fitness trackers. This is why we see *wearables* as functionally linked by the function of being wearable, while *clothes* tend to be more functionally adjacent in complementing each other to protect the wearer from the weather – although the strength of both the common link and adjacency are clearly a matter of degree. *Belongings* are similar to this group, but are linked through a possessor relationship.

Third, in section 3.1 we already noted some body parts conceived of as pluralities with non-individuated parts, such as *bowels*. For some, these have a metaphorical functional extension: one can *have balls* ‘be audacious’, *brains* ‘be smart’, or *guts* ‘be courageous’. We see these as linked to the same property (of audacity, intelligence, etc.), and this is also how *having (good) manners*, *smarts* and *wits* (which are not based on a body part) can be analysed. Similarly, *(facial) features* and *looks* are linked to a single person, whom they are spatially part of and whose appearance they define in functional terms.

Fourth, some terms for social interactions surface as plurals: *amends*, *congratulations*, *regards*, *respects*, and *thanks*. It is noteworthy that these imply some sort of deference and/or belong to a more polite register. People *give thanks* only to deities and *pay their respects* at a funeral. To *make amends* is more formal than *saying sorry* and *congratulations* is more formal than *happy birthday*. On some level, an act like *giving thanks* is conceived of as having subevents (making an offering, saying a prayer, ...) that are linked through the function of thanking someone. Acknowledging the internal plurality renders the act

¹⁵This topic has spawned a vast literature (see e.g. Quirk et al. 1985: 316, 771; Corbett 2006: 211–213). Typically, nouns like *committee* trigger plural agreement when the emphasis is on the differentiation of members acting individually or in disagreement, while the singular is used when it is viewed as a single, unified body (Depraetere 2003). Interestingly, *committee*-type nouns can also be seen as (functionally connected) clusters (cf. Wagiel 2021 on Slavic collectives), and the factors affecting agreement are the same.

more important, thereby reflecting the social status of the beneficiary – an effect we can also see in some other uncountable plurals (e.g. *ashes* and *remains*, which are reserved for the remains of humans and cities and the like), and which can be linked to the use of plural pronouns for polite 2sg address in languages like French.

Finally, there are a number of smaller groups of nouns denoting functionally connected clusters, of which we can only discuss a small number here.¹⁶ *Tariffs*, *tolls*, and *levies* are functionally overlapping in their function of taxing various goods. *Goods*, *supplies*, *victuals/vittles*, and *provisions* are functionally overlapping in their uses for consumers. *Rights* form a coherent (functionally adjacent) set resulting from an entity’s status (as an animal, human, etc. – providing common linkage). *Notes (of peach)* are all ways in which the abstract idea of peach manifests itself, and are thus functionally linked to that same abstract idea.¹⁷

To finish our discussion of the functional domain, note that functional sequentiality appears to be very rare. One could argue that the parts making up a ceremony such as *banns* or *nuptials* are meaningfully ordered and therefore not only temporally but also functionally sequentially adjacent, each part being the logical next step for the previous part. However, we do not see any clear instances where sequential functional connectedness does not piggy-back on sequentiality in the temporal domain.

3.4 Summary

In summary, the wide semantic variety of English uncountable plurals can be accurately accounted for using the mereotopological notions of overlap (Strong Connectedness), adjacency (External Connectedness), and common linkage, when these notions are applied to the spatial, temporal, and functional domains.

Although overlap, adjacency, and common linkage can be defined precisely, there are uncountable plurals that can be analysed as displaying either overlap or adjacency, for instance. The degree to which their parts are conceptualised as overlapping may differ from context to context. Furthermore, pluralities may be clustered together in multiple interacting ways, as in spatio-temporal *directions* or spatio-political *United States*. Our aim is therefore not to provide an algorithm for classifying plurals as clustered by any one connectedness relation, so long as it is clear that they can all be analysed as clustered.

To reiterate, as far as we can see, the type of connectedness relation (including sequentiality) and its domain do not have a grammatical reflex. Instead, any kind of cluster linked together by these relations is treated as uncountable.

4 Deriving uncountability

Having shown that uncountable plurals can be analysed as clusters connected through different connectedness relations, we now address the question why the parts of clusters cannot be counted using modification by numerals. It is important to remember that this is a grammatical, rather than a cognitive, restriction: it is perfectly possible to count and talk about *five news items* and *the number of blades in a pair of scissors*, but this requires more syntactic structure.

This renders the clusters denoted by uncountable plurals similar to MSSC entities like *book*. Trivially, the parts of MSSC entities cannot be counted using numeral modification (**fifty book* to mean ‘a book of fifty pages’), even though we can refer to them by other means (*a book of fifty pages*). The MSSC property provides a sensible formalisation of the conceptual notion of a “whole”, and it seems intuitive that for wholes speakers need to do more work to access their parts than to count the wholes themselves

¹⁶The supplementary material contains a list of the uncountable plurals in our data set and their classification.

¹⁷Note that liquors, for instance, may have *notes of peach* even when no peach was used in their production, and that *note* can therefore not be taken as denoting a portion of material. We analyse these classifier constructions with abstract complements in more detail in Rooryck & Staps (2025).

(which is possible with numeral modification: *fifty books*). Crucial to the notion of wholes is the fact that MSSC entities are maximal: two thirds of a sphere are an SSC entity, but they are not an MSSC entity and do not make up the whole sphere.

What we would like to suggest, therefore, is that uncountable plurals denote *maximal* clusters. This seems appropriate for the examples we have seen so far (e.g., part of the *Netherlands* is only *Netherlands* in the same universal grinder sense that part of a *chair* is *chair*). Maximality is the formal counterpart of the intuitive notion of wholes. Wholes do not allow direct access to their parts because that is not the level at which they are conceptualised: just as the concept of a *chair* does not immediately bring to mind an individuated seat, backrest, and legs, so the concept of *clothes* does not immediately bring to mind individuated shirts and sweaters. This explains why uncountable plurals cannot be modified by numerals to count their parts.

Formally, maximality can be captured as in (13). In words, $\text{Maximal}(P, C)$ is true of P -entities x that are maximal with respect to a connectedness relation C : if part of another P -entity y is C -related to any part of x , y must be part of x . When C is mereological parthood and the domain is limited to SSC entities, this is equivalent to the MSSC condition, as shown in (14).¹⁸ But when the domain is limited to C -connected clusters, this yields a suitable definition of maximal clusters, as in (15).¹⁹

$$(13) \quad \text{Maximal}(P, C)(x) =_{\text{def}} P(x) \wedge \forall y [(P(y) \wedge \exists a \leq x, b \leq y [C(a, b)]) \leftrightarrow y \leq x]$$

$$(14) \quad \text{MSSC}(P)(x) \equiv \text{Maximal}(P \wedge \text{SSC}, \leq)(x)$$

$$(15) \quad \text{MCluster}(P, C)(x) =_{\text{def}} \text{Maximal}(\text{Cluster}(P, C), C)(x)$$

The conclusion, then, is that uncountable plurals are in fact countable – but they are counted at the level of the cluster rather than that of the parts. This is why we can perfectly talk about pluralities of clusters, with a classifier construction (*five pairs of scissors*; see Corbett 2019: 92–94 for more general discussion), but often also using a zero plural (*five headquarters*).²⁰

5 Conclusions

Previous work on uncountable plurals, in English and beyond, has often focused on specific semantic fields, such as “mass plurals” (e.g. *clothes*, *resources*; Acquaviva 2004) and “aggregates” (e.g. *collectibles*; Gardelle 2016). Pluralia tantum are commonly subdivided into descriptive categories such as “bipartites”, “internally complex events”, “diseases and ailments”, etc. (Acquaviva & Gardelle 2023, see also Koptjevskaja-Tamm & Wälchli 2001: 630). Fully recognising of the value of these descriptions, our aim here has been to show that such categories are related to each other on a more abstract level. In particular, they can all be analysed as clusters in the mereotopological sense.

Our focus has been on nouns that show morphological signs of denoting pluralities. Group and swarm nouns like *committee* are interesting for similar reasons: they show morphosyntactic signs of denoting pluralities (optionally controlling plural agreement), even though this is not reflected in morphology. These denote spatial, temporal, and “non-spatiotemporal” clusters (Henderson 2017), which suggests that

¹⁸After rewriting, this boils down to equivalence between $O(x, y) = \exists z [z \leq x \wedge z \leq y]$ and $\exists a \leq x, b \leq y [a \leq b]$. First, if $O(x, y)$, take $a = b = z$ so that trivially $a \leq b$. Second, given $a \leq x, b \leq y$ with $a \leq b$, take $z = a$ with $z \leq y$ due to transitivity of $\leq$.

¹⁹Although maximality is a binary property, the degree to which parts are accessible – and by extension uncountability – is a matter of degree. The president of the United States, unlike the first lady, represents the union of states and interacts with the governments of the individual states. Correspondingly we find examples of *the President of the 50 United States* but cannot count the parts of the plurality in *the First Lady of the (*50) United States*. This does not detract from the fact that the *United States* form a maximal cluster, but shows that the unity of the cluster can be conceptualised as more or less strong depending on the context.

²⁰It is true that there are uncountable plurals that cannot be pluralised further (e.g., *metaphysics* can only be pluralised after being lifted to a kind). However, the same is true for regular singulars (e.g., *happiness*), so this fact does not require a separate explanation here.

uncountable plurals are part of a broader phenomenon whereby clusters have some but not all of the morphologically and morphosyntactically marked features of plain sums. Interestingly, clusters of animate entities are always lexicalised as group or swarm nouns and never as uncountable plurals, and inanimate swarm nouns (there are no inanimate group nouns) do not trigger plural agreement (de Vries 2015: ch. 6; Wągiel 2021: 179). We leave the relations between semantics and morphological form (with or without plural -s) as well as morphosyntactic properties such as agreement, modification by *some* and *many*, and modification by the indefinite article (*a headquarters/*suburbs*) for further study.

Our analysis contributes to the broader framework of mereotopology in three ways. First, it joins a growing body of research describing the domains in which metaphorical extensions of the basic spatial notions can be applied. In view of the variety of semantic types in especially the functional domain, we have not formalised our descriptions in the non-spatial domains, as Mazzola (2019), Wągiel (2024), and McNally (2024) have done for events and Wągiel (2021) for social roles (in our approach a subset of functional relations). We hope that given this earlier work it is clear how mereotopological notions can be extended to the required domains.

Second, we have discussed different connectedness relations. On the one hand, we have suggested in passing that the notion of Proximate Connectedness may not have the same primitive status as Strong Connectedness (overlap) and External Connectedness (adjacency). Proximate Connectedness was introduced by Grimm (2012) to account for clusters like swarms. However, we believe that these are better accounted for using adjacency in the functional domain, which can capture family relationships and the like (this also allows the theory to capture two swarms flying through each other while maintaining their distinct identity, which a purely spatial analysis would not be able to do). On the other hand, we introduced a new connectedness relation – common linkage – to capture clusters of parts linked together through a common event or entity, often a “root” or “origin”.

Third, we have introduced maximality as a property of clusters. This provides a parallel between MSSC entities and maximal clusters, which can account for similarities in their behaviour – in our case, the fact that their parts cannot be accessed directly.

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